

SAE 303

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Maquette :

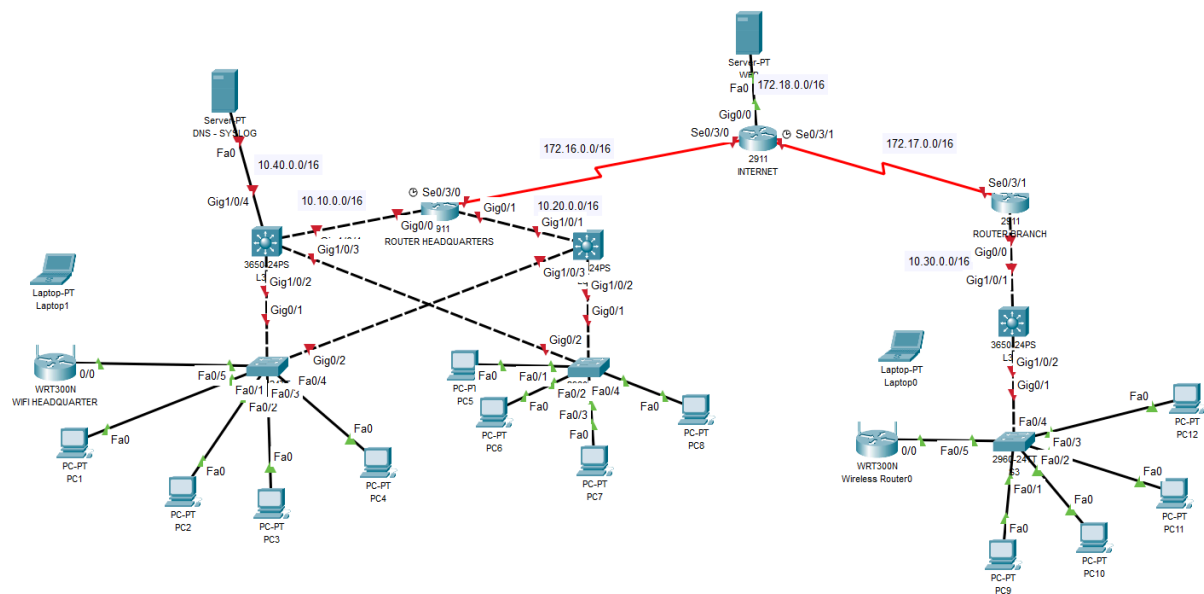


Table D'adressage :

Équipements

NOM	PORT	IP	DIRECTION
INTERNET	Se0/3/0	172.16.0.1/16	ROUTER HEADQUARTERS
	Se0/3/1	172.17.0.1/16	ROUTER BRANCH
	Gig0/0	172.18.0.1/16	WEB
ROUTER HEADQUARTERS	Se0/3/0	172.16.0.2/16	INTERNET
	Gig0/0	10.10.0.1/16	L3S1
	Gig0/1	10.20.0.1/16	L3S2
ROUTER BRANCH	Se0/3/1	172.17.0.2/16	INTERNET

	Gig0/0	10.30.0.1/16	L3S3
L3S1	Gig1/0/1	10.10.0.2/16	ROUTER HEADQUARTERS
	Gig1/0/2	- - - - -	S1
	Gig1/0/3	- - - - -	S2
	Gig1/0/4	10.40.0.1/16	DHCP-DNS-SYSLOG
L3S2	Gig1/0/1	10.20.0.2/16	ROUTER HEADQUARTERS
	Gig1/0/2	- - - - -	S2
	Gig1/0/3	- - - - -	S1
L3S3	Gig1/0/1	10.30.0.2/16	ROUTER BRANCH
	Gig1/0/2	- - - - -	S3
S1	Gig0/1	- - - - -	L3S1
	Gig0/2	- - - - -	L3S2
	Fa0/1	- - - - -	PC1
	Fa0/2	- - - - -	PC2
	Fa0/3	- - - - -	PC3
	Fa0/4	- - - - -	PC4
	Fa0/5	- - - - -	WIFI-INTERNE- HEAD
	VLAN 10	192.168.10.4/24	- - - - -
S2	Gig0/1	- - - - -	L3S2
	Gig0/2	- - - - -	L3S1
	Fa0/1	- - - - -	PC5
	Fa0/2	- - - - -	PC6
	Fa0/3	- - - - -	PC7
	Fa0/4	- - - - -	PC8
	Fa0/5	- - - - -	WIFI-INVITE-HEAD
	VLAN 10	192.168.10.5/24	- - - - -
S3	Gig0/1	- - - - -	L3S3
	Fa0/1	- - - - -	PC9
	Fa0/2	- - - - -	PC10
	Fa0/3	- - - - -	PC11

	Fa0/4	- - - - -	PC12
	Fa0/5	- - - - -	WIFI-INTERNE-BRANCH
	Fa0/6	- - - - -	WIFI-INVITE-BRANCH
	VLAN 10	192.168.110.6/24	- - - - -
WEB	Fa0	172.18.0.2/16	INTERNET
DHCP-DNS-SYSLOG	Fa0	10.40.0.2/16	L3S1

VLAN

NOM	IDENTIFIANT	IP
ADMIN-HEAD	10	192.168.10.0/24
PERSONNEL-HEAD	20	192.168.20.0/24
PRODUCTION-HEAD	30	192.168.30.0/24
VIDEO-HEAD	40	192.168.40.0/24
WIFI-INTERNE-HEAD	50	192.168.50.0/24
WIFI-INVITE-HEAD	60	192.168.60.0/24
ADMIN-BRANCH	10	192.168.110.0/24
PERSONNEL-BRANCH	20	192.168.120.0/24
PRODUCTION-BRANCH	30	192.168.130.0/24
VIDEO-BRANCH	40	192.168.140.0/24
WIFI-INTERNE-BRANCH	50	192.168.150.0/24
WIFI-INVITE-BRANCH	60	192.168.160.0/24

Tunnel

INTERFACE	IP	DIRECTION
HEADQUARTERS Tunnel 0	1.1.1.1/16	BRANCH
BRANCH Tunnel 0	1.1.1.2/16	HEADQUARTERS

Configuration :

INTERNET :

```

en
conf t
!
service password-encryption
!
hostname INTERNET
!
ip domain-name SAE303.local
!
enable secret proctr00
!
username admin privilege 15 secret proctr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
interface Gig0/0
description WEB
ip address 172.18.0.1 255.255.0.0
no sh
!
interface Se0/3/0
description ROUTER_HEADQUARTERS
ip address 172.16.0.1 255.255.0.0
no sh
!
interface Se0/3/1
description ROUTER_BRANCH
ip address 172.17.0.1 255.255.0.0
clock rate 128000
no sh
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 172.16.0.0 0.0.255.255 area 0
network 172.17.0.0 0.0.255.255 area 0
network 172.18.0.0 0.0.255.255 area 0
!
line con 0
!
logging synchronous
login local
exec-timeout 10 0
!
line aux 0
no exec
!
line vty 0 4
transport input ssh

```

```

logging synchronous
login local
exec-timeout 10 0
!
ip route 0.0.0.0 0.0.0.0 172.16.0.2
!
end

```

ROUTER HEADQUARTERS:

```

en
conf t
license boot module c2900 technology-package securityk9
reload

```

```

en
conf t
!
hostname ROUTER_HEADQUARTERS
service password-encryption
enable secret progtr00
ip domain-name SAE303.local
!
username admin privilege 15 secret progtr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
ip default-gateway 192.168.10.254
!
logging host 10.40.0.2
logging trap debugging
!
class-map match-any VIDEO_TRAFFIC
match dscp af41
!
policy-map QOS_OUTPUT
class VIDEO_TRAFFIC
priority percent 20
class class-default
fair-queue
!
access-list 10 remark ACCES_SSH_ADMIN_UNIQUEMENT
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.110.0 0.0.0.255
access-list 10 deny any
!
access-list 101 permit gre host 172.16.0.2 host 172.17.0.2
access-list 101 permit esp host 172.16.0.2 host 172.17.0.2

```

```

access-list 101 permit udp host 172.16.0.2 host 172.17.0.2 eq isakmp
!
access-list 110 deny ip 192.168.0.0 0.0.255.255 10.30.0.0 0.0.255.255
access-list 110 deny ip any 10.30.0.0 0.0.255.255
access-list 110 deny ip any 10.40.0.0 0.0.255.255
access-list 110 permit ip 192.168.0.0 0.0.255.255 any
access-list 110 permit ip 10.0.0.0 0.255.255.255 any
!
crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
!
crypto isakmp key progtr00 address 172.17.0.2
!
crypto ipsec transform-set myset esp-aes esp-sha-hmac
!
crypto map MYMAP 10 ipsec-isakmp
set peer 172.17.0.2
set transform-set myset
match address 101
!
interface GigabitEthernet0/0
description VERS_L3S1
ip address 10.10.0.1 255.255.0.0
ip nat inside
no sh
!
interface GigabitEthernet0/1
description VERS_L3S2
ip address 10.20.0.1 255.255.0.0
ip nat inside
no sh
!
interface Serial0/3/0
description VERS_INTERNET
ip address 172.16.0.2 255.255.0.0
ip nat outside
clock rate 128000
crypto map MYMAP
service-policy output QOS_OUTPUT
no sh
!
interface Tunnel0
ip address 1.1.1.1 255.255.255.252
tunnel source Serial0/3/0
tunnel destination 172.17.0.2
no shutdown
!
ip nat inside source list 110 interface Serial0/3/0 overload
!
ip route 0.0.0.0 0.0.0.0 172.16.0.1
ip route 172.17.0.2 255.255.255.255 172.16.0.1
ip route 10.30.0.0 255.255.0.0 1.1.1.2
ip route 192.168.0.0 255.255.0.0 10.10.0.2
ip route 192.168.0.0 255.255.0.0 10.20.0.2

```

```

ip route 10.40.0.0 255.255.0.0 10.10.0.2
ip route 192.168.110.0 255.255.255.0 1.1.1.2
ip route 192.168.120.0 255.255.255.0 1.1.1.2
ip route 192.168.130.0 255.255.255.0 1.1.1.2
ip route 192.168.140.0 255.255.255.0 1.1.1.2
ip route 192.168.150.0 255.255.255.0 1.1.1.2
ip route 192.168.160.0 255.255.255.0 1.1.1.2
!
line vty 0 4
transport input ssh
login local
access-class 10 in
logging synchronous
exec-timeout 10 0
end
wr

```

ROUTER BRANCH :

```

en
conf t
!
hostname ROUTER_BRANCH
!
service password-encryption
enable secret progtr00
!
ip domain-name SAE303.local
!
username admin privilege 15 secret progtr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
ip default-gateway 192.168.10.254
!
logging host 10.40.0.2
logging trap debugging
!
interface Gig0/0
description VERS_L3S3_BRANCH
ip address 10.30.0.1 255.255.0.0
ip nat inside
no shutdown
!
interface Serial0/3/1
description VERS_INTERNET
ip address 172.17.0.2 255.255.0.0
ip nat outside

```

```

clock rate 128000
no shutdown
!
crypto isakmp policy 10
  encryption aes
  hash sha
  authentication pre-share
  group 2
!
crypto isakmp key progtr00 address 172.16.0.2
!
crypto ipsec transform-set myset esp-aes esp-sha-hmac
!
access-list 10 remark ACCES_SSH_ADMIN_UNIQUEMENT
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.110.0 0.0.0.255
access-list 10 deny any
!
access-list 101 permit gre host 172.17.0.2 host 172.16.0.2
access-list 101 permit esp host 172.17.0.2 host 172.16.0.2
access-list 101 permit udp host 172.17.0.2 host 172.16.0.2 eq isakmp
!
access-list 110 deny ip 10.30.0.0 0.0.255.255 192.168.0.0 0.0.255.255
access-list 110 deny ip any 10.40.0.0 0.0.255.255
access-list 110 permit ip 10.30.0.0 0.0.255.255 any
access-list 110 permit ip 192.168.0.0 0.0.255.255 any
!
ip access-list extended RESTRICTION_BRANCH
deny ip 192.168.130.0 0.0.0.255 192.168.120.0 0.0.0.255
deny ip 192.168.160.0 0.0.0.255 192.168.0.0 0.0.255.255
deny ip 192.168.160.0 0.0.0.255 10.0.0.0 0.255.255.255
permit ip any any
!
crypto map MYMAP 10 ipsec-isakmp
  set peer 172.16.0.2
  set transform-set myset
  match address 101
!
interface Serial0/3/1
  crypto map MYMAP
!
interface tunnel0
  ip address 1.1.1.2 255.255.255.252
  tunnel source Serial0/3/1
  tunnel destination 172.16.0.2
  no shutdown
!
ip nat inside source list 110 interface Serial0/3/1 overload
!
ip route 172.16.0.2 255.255.255.255 172.17.0.1
ip route 0.0.0.0 0.0.0.0 1.1.1.1
ip route 10.10.0.0 255.255.0.0 1.1.1.1
ip route 10.20.0.0 255.255.0.0 1.1.1.1
ip route 192.168.0.0 255.255.0.0 1.1.1.1
ip route 10.40.0.0 255.255.0.0 1.1.1.1
ip route 192.168.110.0 255.255.255.0 10.30.0.2
ip route 192.168.120.0 255.255.255.0 10.30.0.2

```



```

ip route 192.168.130.0 255.255.255.0 10.30.0.2
ip route 192.168.140.0 255.255.255.0 10.30.0.2
ip route 192.168.150.0 255.255.255.0 10.30.0.2
ip route 192.168.160.0 255.255.255.0 10.30.0.2
!
line vty 0 4
  transport input ssh
  login local
!
end

```

L3S1:

```

en
conf t
!
hostname L3S1
service password-encryption
ip routing
ip domain-name SAE303.local
enable secret progtr00
username admin privilege 15 secret progtr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
ip default-gateway 192.168.10.254
!
logging host 10.40.0.2
logging trap debugging
!
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 deny any
!
line vty 0 15
  login local
  transport input ssh
  access-class 10 in
exit
!
access-list 140 permit ip 192.168.40.0 0.0.0.255 any
access-list 140 permit ip any 192.168.40.0 0.0.0.255
!
ip access-list extended RESTRICTION_SIEGE
deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.60.0 0.0.0.255 192.168.0.0 0.0.255.255
deny ip 192.168.60.0 0.0.0.255 10.0.0.0 0.255.255.255
permit ip any any
!

```

```

class-map match-all VIDEO-TRAFFIC
match access-group 140
exit
!
policy-map PRIORITY-VIDEO
class VIDEO-TRAFFIC
set dscp af41
priority percent 20
exit
!
vlan 1
name SERVER
vlan 10
name ADMIN
vlan 20
name PERSONNEL
vlan 30
name PRODUCTION
vlan 40
name VIDEO
vlan 50
name WIFI-INTERNE
vlan 60
name WIFI-INVITE
exit
!
int vlan10
description ADMIN
ip address 192.168.10.252 255.255.255.0
ip helper-address 10.40.0.2
standby 1 ip 192.168.10.254
standby 1 priority 150
standby 1 preempt
no shutdown
!
int vlan20
description PERSONNEL
ip address 192.168.20.252 255.255.255.0
ip helper-address 10.40.0.2
standby 1 ip 192.168.20.254
standby 1 priority 150
standby 1 preempt
no shutdown
!
int vlan30
description PRODUCTION
ip address 192.168.30.252 255.255.255.0
ip helper-address 10.40.0.2
ip access-group RESTRICTION_SIEGE in
standby 1 ip 192.168.30.254
standby 1 priority 150
standby 1 preempt
no shutdown
!
int vlan40
description VIDEO
ip address 192.168.40.252 255.255.255.0

```

```

ip helper-address 10.40.0.2
standby 1 ip 192.168.40.254
standby 1 priority 150
standby 1 preempt
no shutdown
!
int vlan50
description WIFI-INTERNE
ip address 192.168.50.252 255.255.255.0
ip helper-address 10.40.0.2
standby 1 ip 192.168.50.254
standby 1 priority 150
standby 1 preempt
no shutdown
!
int vlan60
description WIFI-INVITE
ip address 192.168.60.252 255.255.255.0
ip helper-address 10.40.0.2
ip access-group RESTRICTION_SIEGE in
standby 1 ip 192.168.60.254
standby 1 priority 150
standby 1 preempt
no shutdown
!
spanning-tree vlan 10,20,30,40,50,60 root primary
!
int Gig1/0/1
description VERS_ROUTER_HQ
no switchport
ip address 10.10.0.2 255.255.0.0
service-policy output PRIORITY-VIDEO
no shutdown
!
int range Gig1/0/2 - 3
description TRUNK
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60
no shutdown
!
int Gig1/0/4
description VERS_DHCP-DNS-SYSLOG
no switchport
ip address 10.40.0.1 255.255.0.0
no sh
!
ip route 0.0.0.0 0.0.0.0 10.10.0.1
!
end

```

L3S2 :

```

en
conf t

```

```

!
hostname L3S2
service password-encryption
ip routing
ip domain-name SAE303.local
enable secret progtr00
username admin privilege 15 secret progtr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
ip default-gateway 192.168.10.254
!
logging host 10.40.0.2
logging trap debugging
!
access-list 10 remark ACCES_SSH_ADMIN_UNIQUEMENT
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.110.0 0.0.0.255
access-list 10 deny any
!
line vty 0 15
login local
transport input ssh
access-class 10 in
exit
!
access-list 140 permit ip 192.168.40.0 0.0.0.255 any
access-list 140 permit ip any 192.168.40.0 0.0.0.255
!
ip access-list extended RESTRICTION_SIEGE
deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.60.0 0.0.0.255 192.168.0.0 0.0.255.255
deny ip 192.168.60.0 0.0.0.255 10.0.0.0 0.255.255.255
permit ip any any
!
class-map match-all VIDEO-TRAFFIC
match access-group 140
exit
!
policy-map PRIORITY-VIDEO
class VIDEO-TRAFFIC
priority percent 20
exit
!
vlan 10
name ADMIN
vlan 20
name PERSONNEL
vlan 30
name PRODUCTION
vlan 40
name VIDEO
vlan 50

```

```

name WIFI-INTERNE
vlan 60
name WIFI-INVITE
exit
!
int vlan10
description ADMIN
ip address 192.168.10.253 255.255.255.0
ip helper-address 10.40.0.2
standby 1 ip 192.168.10.254
standby 1 priority 100
standby 1 preempt
no shutdown
exit
!
int vlan20
description PERSONNEL
ip address 192.168.20.253 255.255.255.0
ip helper-address 10.40.0.2
standby 1 ip 192.168.20.254
standby 1 priority 100
standby 1 preempt
no shutdown
exit
!
int vlan30
description PRODUCTION
ip address 192.168.30.253 255.255.255.0
ip helper-address 10.40.0.2
ip access-group RESTRICTION_SIEGE in
standby 1 ip 192.168.30.254
standby 1 priority 100
standby 1 preempt
no shutdown
exit
!
int vlan40
description VIDEO
ip address 192.168.40.253 255.255.255.0
ip helper-address 10.40.0.2
standby 1 ip 192.168.40.254
standby 1 priority 100
standby 1 preempt
no shutdown
exit
!
int vlan50
description WIFI-INTERNE
ip address 192.168.50.253 255.255.255.0
ip helper-address 10.40.0.2
standby 1 ip 192.168.50.254
standby 1 priority 100
standby 1 preempt
no shutdown
exit
!
int vlan60

```

```

description WIFI-INVITE
ip address 192.168.60.253 255.255.255.0
ip helper-address 10.40.0.2
ip access-group RESTRICTION_SIEGE in
standby 1 ip 192.168.60.254
standby 1 priority 100
standby 1 preempt
no shutdown
exit
!
spanning-tree vlan 10,20,30,40,50,60 root secondary
!
int Gig1/0/1
description VERS_ROUTER_HQ
no switchport
ip address 10.20.0.2 255.255.0.0
service-policy output PRIORITY-VIDEO
no shutdown
exit
!
int range Gig1/0/2 - 3
description TRUNK
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60
no shutdown
exit
!
ip route 0.0.0.0 0.0.0.0 10.20.0.1
!
end

```

L3S3 :

```

en
conf t
!
hostname L3S3
service password-encryption
ip routing
ip domain-name SAE303.local
username admin privilege 15 secret progtr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
ip default-gateway 192.168.10.254
!
logging host 10.40.0.2
logging trap debugging

```

```

!
access-list 10 remark ACCES_SSH_ADMIN_UNIQUEMENT
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.110.0 0.0.0.255
access-list 10 deny any
!
line vty 0 15
login local
transport input ssh
access-class 10 in
!
access-list 140 permit ip 192.168.40.0 0.0.0.255 any
access-list 140 permit ip any 192.168.40.0 0.0.0.255
!
ip access-list extended RESTRICTION_BRANCH_S3
deny ip 192.168.130.0 0.0.0.255 192.168.120.0 0.0.0.255
deny ip 192.168.160.0 0.0.0.255 192.168.0.0 0.0.255.255
deny ip 192.168.160.0 0.0.0.255 10.0.0.0 0.255.255.255
permit ip any any
!
class-map match-all VIDEO-TRAFFIC
match access-group 140
!
policy-map PRIORITY-VIDEO
class VIDEO-TRAFFIC
priority percent 20
!
vlan 10
name ADMIN
vlan 20
name PERSONNEL
vlan 30
name PRODUCTION
vlan 40
name VIDEO
vlan 50
name WIFI-INTERNE
vlan 60
name WIFI-INVITE
!
int vlan10
description ADMIN
ip address 192.168.110.254 255.255.255.0
ip helper-address 10.40.0.2
no shutdown
!
int vlan20
description PERSONNEL
ip address 192.168.120.254 255.255.255.0
ip helper-address 10.40.0.2
no shutdown
!
int vlan30
description PRODUCTION
ip address 192.168.130.254 255.255.255.0
ip helper-address 10.40.0.2
ip access-group RESTRICTION_BRANCH_S3 in

```

```

no shutdown
!
int vlan40
description VIDEO
ip address 192.168.140.254 255.255.255.0
ip helper-address 10.40.0.2
no shutdown
!
int vlan50
description WIFI-INTERNE
ip address 192.168.150.254 255.255.255.0
ip helper-address 10.40.0.2
no shutdown
!
int vlan60
description WIFI-INVITE
ip address 192.168.160.254 255.255.255.0
ip helper-address 10.40.0.2
ip access-group RESTRICTION_BRANCH_S3 in
no shutdown
!
int Gig1/0/1
description VERS_ROUTER_HQ
no switchport
ip address 10.30.0.2 255.255.0.0
service-policy output PRIORITY-VIDEO
no shutdown
! Adaptez le range selon vos branchements réels (ex: Gig1/0/2 à Gig1/0/4)
int Gig1/0/2
description TRUNK
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60
no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.30.0.1
!
end

```

S1:

```

en
!
conf t
!
hostname S1
!
enable secret progtr00
service password-encryption
!
ip domain-name SAE303.local
username admin privilege 15 secret progtr00
!
crypto key generate rsa
!

```



```

ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
line vty 0 15
login local
transport input ssh
access-class 10 in
!
line console 0
login local
!
ip default-gateway 192.168.10.254
!
logging host 10.40.0.2
logging trap debugging
!
access-list 10 remark ACCES_SSH_ADMIN_UNIQUEMENT
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.110.0 0.0.0.255
access-list 10 deny any
!
vlan 10
name ADMIN
!
vlan 20
name PERSONNEL
!
vlan 30
name PRODUCTION
!
vlan 40
name VIDEO
!
vlan 50
name WIFI-INTERNE
!
vlan 60
name WIFI-INVITE
!
int vlan 10
ip address 192.168.10.4 255.255.255.0
no sh
!
int fa0/1
switchport mode access
switchport access vlan 10
no sh
!
int fa0/2
switchport mode access
switchport access vlan 20
no sh
!
int fa0/3
switchport mode access
switchport access vlan 30

```

```

no sh
!
int fa0/4
switchport mode access
switchport access vlan 40
no sh
!
int fa0/5
switchport mode access
switchport access vlan 50
no sh
!
int range Gig0/1-2
switchport mode trunk
switchport trunk allowed vlan 10-60
no sh
!
int range fa0/6-24
shutdown
!
end

```

S2 :

```

en
!
conf t
!
hostname S2
!
enable secret proctr00
service password-encryption
!
ip domain-name SAE303.local
username admin privilege 15 secret proctr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
line vty 0 15
login local
transport input ssh
access-class 10 in
!
line console 0
login local
!
ip default-gateway 192.168.10.254
!
logging host 10.40.0.2
logging trap debugging

```

```

!
access-list 10 remark ACCES_SSH_ADMIN_UNIQUEMENT
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.110.0 0.0.0.255
access-list 10 deny any
!
vlan 10
name ADMIN
!
vlan 20
name PERSONNEL
!
vlan 30
name PRODUCTION
!
vlan 40
name VIDEO
!
vlan 50
name WIFI-INTERNE
!
vlan 60
name WIFI-INVITE
!
int vlan 10
ip address 192.168.10.5 255.255.255.0
no sh
!
int fa0/1
switchport mode access
switchport access vlan 10
no sh
!
int fa0/2
switchport mode access
switchport access vlan 20
no sh
!
int fa0/3
switchport mode access
switchport access vlan 30
no sh
!
int fa0/4
switchport mode access
switchport access vlan 40
no sh
!
int fa0/5
switchport mode access
switchport access vlan 60
no sh
!
int range Gig0/1-2
switchport mode trunk
switchport trunk allowed vlan 10-60
!

```

```
int range fa0/6-24
shutdown
!
end
```

S3:

```
en
!
conf t
!
hostname S3
!
enable secret progtr00
service password-encryption
!
ip domain-name SAE303.local
username admin privilege 15 secret progtr00
!
crypto key generate rsa
!
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
line vty 0 15
login local
transport input ssh
access-class 10 in
!
line console 0
login local
!
ip default-gateway 192.168.110.254
!
logging host 10.40.0.2
logging trap debugging
!
access-list 10 remark ACCES_SSH_ADMIN_UNIQUEMENT
access-list 10 permit 192.168.10.0 0.0.0.255
access-list 10 permit 192.168.110.0 0.0.0.255
access-list 10 deny any
!
int vlan 10
ip address 192.168.110.6 255.255.255.0
no sh
!
vlan 10
name ADMIN
!
vlan 20
name PERSONNEL
!
vlan 30
```

```

name PRODUCTION
!
vlan 40
name VIDEO
!
vlan 50
name WIFI-INTERNE
!
vlan 60
name WIFI-INVITE
!
int fa0/1
switchport mode access
switchport access vlan 10
no sh
!
int fa0/2
switchport mode access
switchport access vlan 20
no sh
!
int fa0/3
switchport mode access
switchport access vlan 30
no sh
!
int fa0/4
switchport mode access
switchport access vlan 40
no sh
!
int fa0/5
switchport mode access
switchport access vlan 50
no sh
!
int fa0/6
switchport mode access
switchport access vlan 60
no sh
!
int Gig0/1
switchport mode trunk
switchport trunk allowed vlan 10-60
no sh
!
int range fa0/7-24
shutdown
!
end

```

SAE303-GODART-LAURENT.pkt

SAE303-GODART-LAURENT - Copie.pkt

Étape 1 : Préparation du Laptop (Matériel)

Sur Packet Tracer, les ordinateurs portables ont souvent une carte Ethernet par défaut. Il faut mettre une carte Wifi.

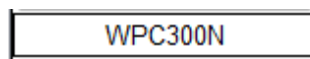
1. Clique sur **Laptop1**.
2. Dans l'onglet **Physical**, éteins le PC (petit bouton rouge sur le côté).



3. Retire le module Ethernet (glisse-le vers la liste de gauche).



1. Insère le module **WPC300N** (module Wifi) dans l'emplacement vide.



1. Rallume le PC



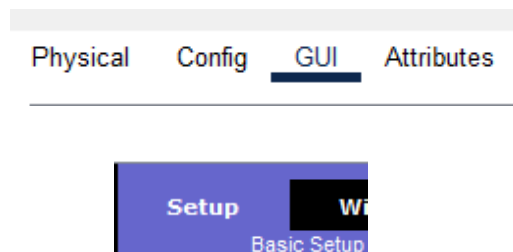
Étape 2 : Configuration du Routeur Wifi (WRT300N)

Le routeur Wifi n'a pas de console CLI comme les switchs Cisco, tout se fait via l'interface graphique (onglet GUI).

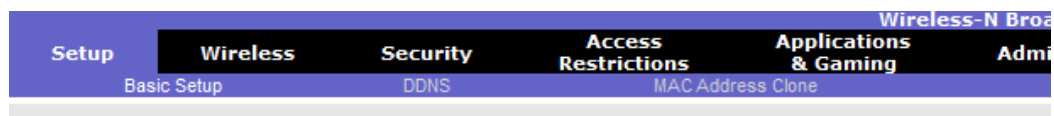
A. Configuration IP (Interface WAN/Internet)

Le câble est branché sur le port "Internet" du WRT300N vers S1.

1. Clique sur le routeur **WRT300N**.
2. Va dans l'onglet **GUI** > Sous-onglet **Setup** > **Basic Setup**.



1. Dans la partie **Internet Setup** (Connection Type), choisis **Static IP** (C'est le plus sûr pour que ça marche tout de suite, car on ignore si le DHCP est parfaitement relayé sur ce port spécifique).
 - **Internet IP Address** : 192.168.50.10 (Une IP libre dans le VLAN 50).
 - **Subnet Mask** : 255.255.255.0
 - **Default Gateway** : 192.168.50.254 (L'adresse virtuelle HSRP de tes switchs L3).
 - **DNS 1** : 192.168.10.10 (L'adresse de ton serveur DNS/Syslog).



Static IP ▼

Internet IP Address:	192	.	168	.	50	.	2
Subnet Mask:	255	.	255	.	255	.	0
Default Gateway:	192	.	168	.	50	.	254
DNS 1:	192	.	168	.	10	.	10
DNS 2 (Optional):	0	.	0	.	0	.	0
DNS 3 (Optional):	0	.	0	.	0	.	0

Host Name:

Domain Name:

MTU: ▼ Size:

IP Address: . . .

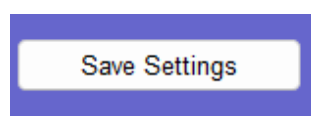
Subnet Mask: ▼

2. Dans la partie **Network Setup** (en bas, côté LAN/Wifi) :

- Laisse l'IP par défaut : **192.168.0.1**.
- Assure-toi que **DHCP Server** est **Enabled**.

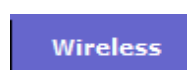
DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

3. Clique sur **Save Settings** (tout en bas).



B. Configuration du Wifi (SSID et Sécurité)

1. Va dans l'onglet **Wireless** > **Basic Wireless Settings**.



Basic Wireless Settings

- **Network Name (SSID) :** `WIFI-HQ-PERSONNEL` (Nom arbitraire).

Setup	Wireless	Security	Access Restrictions	Applications & Gaming	Adminis
Basic Wireless Settings	Wireless Security	Guest Network	Wireless MAC Filter		

Network Mode:	Mixed
Network Name (SSID):	WIFI-HQ-PERSONNEL
Radio Band:	Auto
Wide Channel:	Auto
Standard Channel:	1 - 2.412GHz
SSID Broadcast:	☒ Enabled ☐ Disabled

- Cliquez sur **Save Settings**.

Save Settings

1. Va dans le sous-onglet **Wireless Security**.

Wireless Security

- **Security Mode :** `WPA2 Personal` .
- **Passphrase :** `progr00` (Le mot de passe utilisé partout dans ton projet).

Security Mode:	WPA2 Personal
Encryption:	AES
Passphrase:	progr00
Key Renewal:	3600 seconds

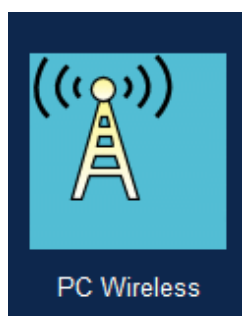
- Cliquez sur **Save Settings**.

Save Settings

Étape 3 : Connecter le Laptop au Wifi

Maintenant que le routeur émet, connectons le PC.

1. Clique sur **Laptop1** > onglet **Desktop** > icône **PC Wireless**.



1. Va dans l'onglet **Connect**.

Connect

1. Attends quelques secondes, le réseau **WIFI-HQ-PERSONNEL** devrait apparaître.

Wireless Network Name	CH	Signal
WIFI-HQ-...	1	100%

2. Sélectionne-le et clique sur **Connect**.

Connect

3. Entre le mot de passe : **progrtr00** et valide.

Security	WPA2-Personal	▼
Pre-shared Key	progrtr00	

1. Ferme la fenêtre "PC Wireless".

Étape 4 : Vérification et Test

1. Ouvre l'invite de commande (**Command Prompt**) sur le **Laptop1**.
2. Tape `ipconfig`. Tu devrais avoir une IP du type `192.168.0.x` (donnée par le WRT300N).
3. Teste la communication vers le serveur DNS/Syslog :

```
ping 192.168.10.10
```

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=30ms TTL=126
Reply from 192.168.10.10: bytes=32 time=28ms TTL=126
```

1. Teste la communication vers le serveur Web (Internet) :

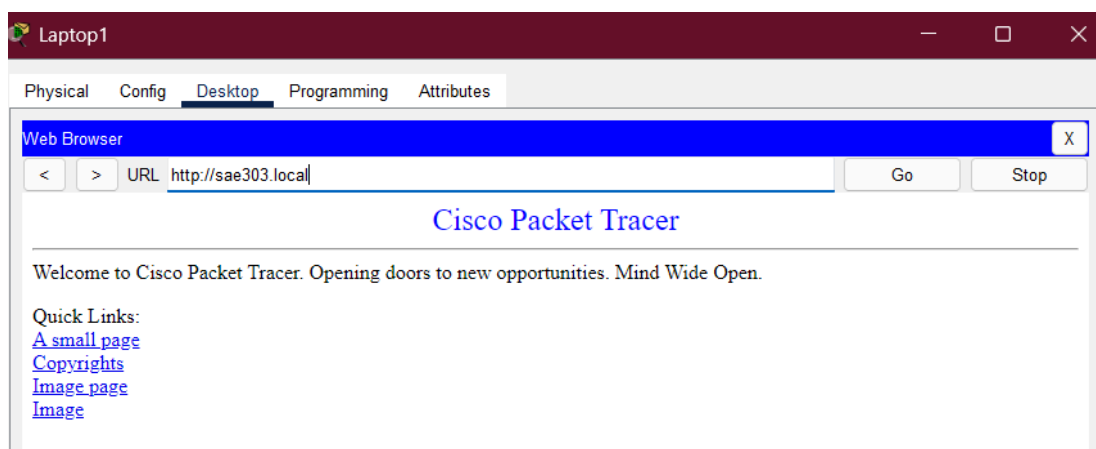
```
ping 172.18.0.2
```

```
C:\>ping 172.18.0.2

Pinging 172.18.0.2 with 32 bytes of data:

Reply from 172.18.0.2: bytes=32 time=17ms TTL=124
Reply from 172.18.0.2: bytes=32 time=29ms TTL=124
Reply from 172.18.0.2: bytes=32 time=32ms TTL=124

Ping statistics for 172.18.0.2:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 32ms, Average = 26ms
```



```
username admin privilege 15 secret progt00
```

```
logging host 10.40.0.2
```

```
logging trap warnings
```

```
service password-encryption
```

```
enable secret progtr00
```

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	
VLAN60-BRANCH	192.168.160.254	10.40.0.2	192.168.160.11	255.255.255.0	225	!
VLAN50-BRANCH	192.168.150.254	10.40.0.2	192.168.150.11	255.255.255.0	225	!
VLAN40-BRANCH	192.168.140.254	10.40.0.2	192.168.140.11	255.255.255.0	225	!
VLAN30-BRANCH	192.168.130.254	10.40.0.2	192.168.130.11	255.255.255.0	225	!
VLAN20-BRANCH	192.168.120.254	10.40.0.2	192.168.120.11	255.255.255.0	225	!
VLAN10-BRANCH	192.168.110.254	10.40.0.2	192.168.110.11	255.255.255.0	225	!
VLAN60-HEAD	192.168.60.254	10.40.0.2	192.168.60.11	255.255.255.0	225	!
VLAN50-HEAD	192.168.50.254	10.40.0.2	192.168.50.11	255.255.255.0	225	!
VLAN40-HEAD	192.168.40.254	10.40.0.2	192.168.40.11	255.255.255.0	225	!
VLAN30-HEAD	192.168.30.254	10.40.0.2	192.168.30.11	255.255.255.0	225	!
VLAN20-HEAD	192.168.20.254	10.40.0.2	192.168.20.11	255.255.255.0	225	!
VLAN10-HEAD	192.168.10.254	10.40.0.2	192.168.10.11	255.255.255.0	225	!

No.	Name	Type	Detail
0	sae303.local	A Record	172.18.0.2

```
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 2
!
line vty 0 15
login local
transport input ssh
```

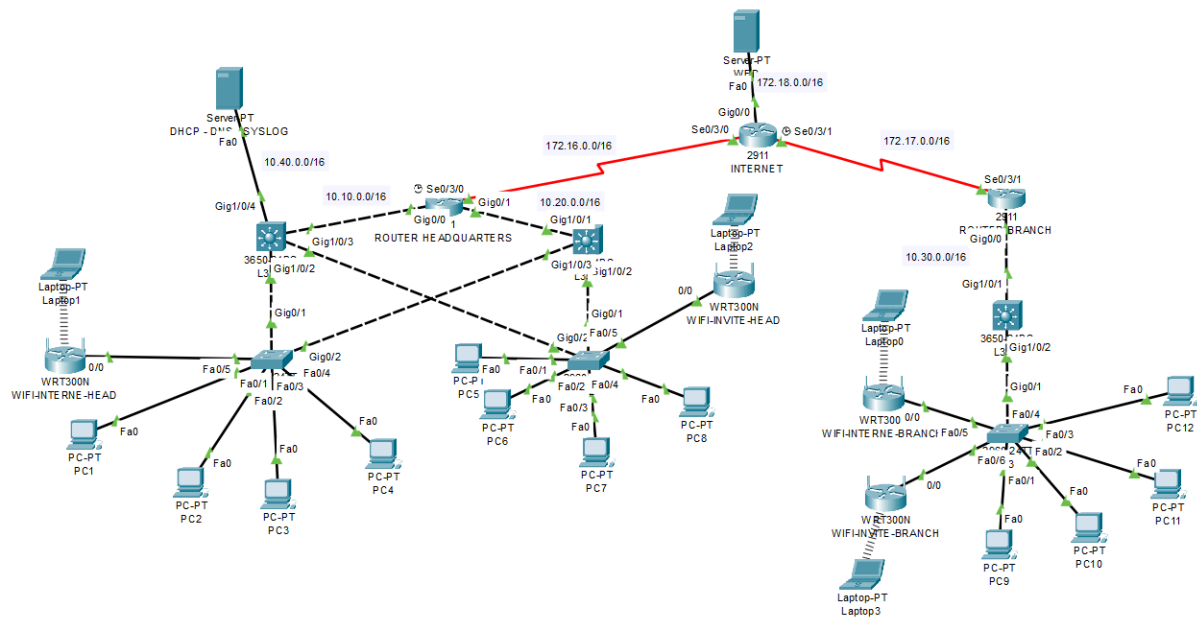
Syslog

Service ☒ On ☐ Off

	Time	HostName	Message
1	-	10.40.0.1	%SYS-5-CONFIG_I: Configur...
2	-	10.40.0.1	%SYS-6-LOGGINGHOST_STARTSTO...
3	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
4	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
5	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
6	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
7	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
8	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
9	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
10	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
11	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
12	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
13	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
14	-	10.40.0.1	%HSRP-6-STATECHANGE: ...
15	-	10.20.0.2	%HSRP-6-STATECHANGE: ...
16	-	10.20.0.2	%HSRP-6-STATECHANGE: ...

SAE303-GODART-LAURENT.pkt

POUR LA PHOTO DU RAPPORT :



SAE303-GODART-LAURENT.pkt

```
ip route 1.1.1.0 255.255.255.0 172.17.0.2
```

SAE303-GODART-LAURENT.pkt

```
ip access-list extended NO_PING_30_40_HEAD
deny icmp 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
deny icmp 192.168.30.0 0.0.0.255 192.168.140.0 0.0.0.255
deny icmp 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
deny icmp 192.168.40.0 0.0.0.255 192.168.130.0 0.0.0.255
interface vlan 30
ip access-group NO_PING_30_40_HEAD in
exit
interface vlan 40
ip access-group NO_PING_30_40_HEAD in
```

```
ip access-list extended NO_PING_30_40_BRANCH
deny icmp 192.168.130.0 0.0.0.255 192.168.140.0 0.0.0.255
deny icmp 192.168.130.0 0.0.0.255 192.168.40.0 0.0.0.255
```

```
deny icmp 192.168.140.0 0.0.0.255 192.168.130.0 0.0.0.255
deny icmp 192.168.140.0 0.0.0.255 192.168.30.0 0.0.0.255
permit ip any any

interface vlan 30
 ip access-group NO_PING_30_40_BRANCH in
 exit
interface vlan 40
 ip access-group NO_PING_30_40_BRANCH in
 exit
```